DeskSense AI

Intelligent & Emotionally Adaptive
Workspace Assistant

PROJECT REPORT

A Workspace That Adapts to You.

Project Category
Artificial Intelligence • Internet of Things • Smart Workspace • Human-Centred Computing

Hardware and Software Integrated Prototype

1. Abstract

Modern employees, students and professionals spend considerable portions of their day working at desks and computer workstations. Prolonged screen exposure, unsuitable lighting, uncomfortable environmental conditions, limited movement and demanding work sessions can negatively affect workspace comfort and the overall working experience.

DeskSense AI is an affordable, AIoT-powered workspace monitoring and assistance system designed to address these challenges. It combines environmental sensors, computer vision, contextual analysis and user feedback to understand workspace conditions and provide personalised recommendations.

Unlike conventional desk-monitoring systems that primarily focus on physical ergonomics, DeskSense AI introduces an emotionally adaptive workspace concept. It aims to help users create environments that support relaxation, concentration, comfort and mindfulness.

The system can recommend appropriate lighting, movement breaks, screen-distance adjustments and environmental improvements. Through optional connected devices, it can also adapt workspace ambiance using lighting and other user-approved controls.

DeskSense AI follows a privacy-first approach, prioritising local processing, voluntary user feedback, transparent recommendations and user-controlled automation.

2. Introduction

Workspaces have a significant influence on how people experience their daily activities. In corporate offices, educational institutions and home-working environments, individuals frequently spend extended periods in front of computer screens.

Common workspace challenges include:

- Insufficient or excessive lighting.
- Incorrect screen positioning and viewing distance.
- Extended periods of sitting.
- Uncomfortable temperature and humidity.
- Lack of movement and regular breaks.
- Environmental distractions during demanding tasks.
- Difficulty creating a comfortable atmosphere for focused work or relaxation.

Most basic workspace monitoring solutions focus on individual measurements. However, a more useful system should consider multiple conditions together and translate them into practical recommendations.

DeskSense AI proposes an intelligent workspace assistant that combines physical workspace monitoring with user-centred well-being assistance.

3. Problem Statement

Existing work environments often lack affordable and personalised systems that can monitor workspace conditions and provide meaningful recommendations based on the user's working context.

Environmental measurements, ergonomic indicators and individual comfort preferences are frequently treated separately. Users may not recognise when their workspace conditions need improvement or may struggle to create an environment appropriate for different activities.

Furthermore, conventional monitoring approaches can raise privacy concerns when cameras or employee-monitoring technologies are involved.

The problem addressed by DeskSense AI is the absence of an affordable, privacy-conscious and context-aware workspace assistance system that connects environmental monitoring, ergonomic guidance and personalised ambiance recommendations.

4. Proposed Solution

DeskSense AI is proposed as an intelligent workspace assistant consisting of hardware sensors, a computer-vision module, a backend processing system and an interactive dashboard.

The system collects information from available sensors and user feedback, analyses the combined information and generates explainable recommendations.

- Physical workspace monitoring.
- Environmental condition monitoring.
- User-reported well-being and comfort assistance.
- Context-aware recommendation generation.
- Adaptive workspace ambiance.
- Privacy-conscious data management.

The system is designed to suggest improvements rather than diagnose medical or psychological conditions.

5. Project Objectives

- Develop an affordable IoT-based workspace monitoring device.
- Monitor lighting, temperature, humidity and desk occupancy.
- Estimate screen distance and relevant posture indicators using computer vision.
- Provide personalised ergonomic and environmental recommendations.
- Introduce Calm, Focus, Refresh and Mindfulness workspace modes.
- Consider voluntary user feedback when generating recommendations.
- Enable user-approved lighting and ambiance adjustments.
- Develop an interactive dashboard for monitoring and configuration.
- Protect user privacy through consent-based monitoring and responsible data handling.

6. Target Users and Real-World Applications

DeskSense AI can be adapted for several environments.

Application	Potential use
Corporate offices	Personal workspace comfort and environmental monitoring
IT companies	Focus sessions, break reminders and ergonomic assistance
Colleges and computer laboratories	Study-space monitoring and lighting recommendations
Libraries	Desk occupancy and environmental monitoring
Work-from-home environments	Personalised workspace assistance

Application	Potential use
Call centres and BPOs	Environmental comfort and configurable break reminders
Co-working spaces	Workspace condition monitoring

Potential adopters include educational institutions, corporate facility-management teams, co-working operators and individual users. The system should be positioned as an employee-centred assistance tool, not an employee-surveillance solution.

7. Core Features

7.1 Physical Workspace Intelligence

Eye Comfort Assistance: Estimate face-to-screen distance using webcam-based computer vision, compare it against the user's preferred range and recommend monitor-position or text-size adjustments.

Position Guard: Identify selected posture-related indicators and provide reminders when configured posture or positioning conditions persist.

Movement Assistance: Track desk occupancy and work-session duration, then recommend short movement breaks after configurable periods.

Light Advisor: Monitor ambient brightness through an LDR sensor, identify low-light conditions and recommend suitable lighting adjustments.

Climate Monitoring: Measure room temperature and humidity, then recommend ventilation or cooling when conditions warrant it.

7.2 Emotional Comfort and Ambiance Intelligence

This module extends DeskSense beyond conventional ergonomic monitoring. Instead of attempting to diagnose emotions from facial expressions, the system uses voluntary user feedback and contextual information to provide appropriate workspace recommendations.

- A user reports feeling stressed during a demanding task; DeskSense offers Calm Mode and optional relaxation guidance.
- A user selects Focus Mode before starting a work session; DeskSense recommends suitable lighting and schedules a configurable break.
- A user reports feeling tired; DeskSense suggests a short break or Refresh Mode.

7.3 Intelligent Recommendations

DeskSense translates collected information into human-readable guidance. Example recommendations:

- “You have been working for 55 minutes without a break. Take a short walk and look away from the screen.”
- “Your screen appears closer than your preferred viewing range. Consider moving it back.”
- “Ambient light is low. Consider turning on your desk lamp.”
- “The room temperature is above your preferred range. Consider improving cooling or ventilation.”
- “You selected Focus Mode. Your focus session is scheduled for 50 minutes.”
- “It is 7 p.m. and you have had extended screen exposure. Would you like to activate Wind-down Mode?”

Each recommendation should explain the information that triggered it.

7.4 Adaptive Workspace Modes

Mode	Purpose and possible actions
Calm Mode	Support a more relaxing workspace; recommend softer preferred lighting, offer optional breathing exercises, suggest a short screen break and optionally provide calming audio.
Focus Mode	Support uninterrupted work; recommend suitable task lighting, offer configurable focus sessions, reduce interruptions through supported desktop integrations and schedule movement reminders.
Refresh Mode	Encourage a change of activity; recommend brighter lighting when appropriate, suggest movement and offer simple refresh activities.
Mindfulness Mode	Provide an optional environment for intentional relaxation; recommend softer lighting and offer guided breathing or user-selected ambient audio.

These modes are configurable recommendations, not treatments for stress, fatigue or other health conditions.

8. Intelligent Recommendation Engine

The recommendation engine is the central intelligence component of DeskSense AI. It combines sensor readings, work-session information and voluntary user feedback to determine which recommendations may be useful.

Example decision scenario

Collected information: Work session: 55 minutes; ambient brightness: low; temperature: above the user's preferred setting; user feedback: feeling stressed; lighting preference: warm lighting.

Generated recommendation: “Consider switching to Calm Mode. Your work session has continued for 55 minutes, the ambient lighting is low, and you reported feeling stressed. DeskSense recommends a short break and your preferred warm lighting.”

Explainability

- The detected condition.
- The relevant sensor or user-feedback information.
- The reason for the recommendation.
- The suggested action.
- An option to accept, reject or customise it.

Initial implementation: The first version will use a rule-based recommendation engine. Machine-learning-based personalisation may be introduced after sufficient representative data has been collected and evaluated with appropriate consent.

9. Adaptive Control

DeskSense can integrate with compatible smart devices to provide user-approved workspace adjustments.

Device	Possible functionality
RGB LED strip	Adjustable colour and brightness

Device	Possible functionality
Smart bulbs	Warm or cool lighting
Smart plugs	Control compatible connected devices
Fans	User-approved cooling control
Smart speakers	Optional ambient audio and voice reminders
Smart blinds	Daylight and glare management
Wearables	Optional activity or heart-rate context with consent

Design principle: Suggest first, automate later. Users can approve specific automation rules and customise when those rules operate. They should always retain manual control.

10. Hardware Requirements

10.1 Currently Available Hardware

Component	Quantity	Purpose
NodeMCU ESP8266	1	Main IoT controller
LDR sensor	1	Ambient brightness measurement
Digital temperature/humidity sensors	2	Environmental monitoring; exact model to be verified
HC-SR04 ultrasonic sensor	1	Proximity and occupancy-related measurements
IR sensor	1	Object or presence detection, depending on module
Vibration sensor	1	Desk vibration detection
SIM800L module	1	Optional cellular communication
LEDs	Available	Basic lighting and alert demonstrations
Buzzer	Available	Audible notifications
Laptop webcam	1	Computer-vision input
Jumper wires	Available	Circuit connections

10.2 Optional Additions

Component	Purpose
RGB LED strip	Adaptive ambiance demonstration
CO_2 sensor	Actual CO_2 measurement for ventilation-related recommendations
Suitable desk lamp	Adjustable workspace illumination
Small speaker	Optional calming audio
Breadboard and suitable resistors	Circuit assembly, if not already available
Compatible smart devices	Advanced adaptive control

The DHT22 measures temperature and humidity, not CO_2. A separate CO_2 sensor is necessary for CO_2-related recommendations. The HC-SR04 measures proximity within its sensing range; it cannot independently provide calibrated face-to-screen distance.

11. Software Requirements

Technology	Purpose
Arduino IDE and C++	ESP8266 programming
Python	Data processing and computer vision
OpenCV	Webcam image processing
MediaPipe	Optional face and landmark tracking
Node.js	Backend runtime
Express.js	REST API development
MongoDB	Store sensor readings, preferences and recommendations
React.js	Frontend development
Tailwind CSS	Dashboard styling
Chart.js	Sensor trends and workspace visualisation
Socket.IO	Real-time dashboard updates
Git and GitHub	Version control

12. Proposed System Architecture

The system is divided into five major layers.

Layer 1 – Data Acquisition

Collects data from ESP8266-connected sensors, laptop webcam, user-selected workspace modes and voluntary user feedback.

Layer 2 – Processing

ESP8266 firmware collects sensor readings. Python processes webcam input locally. Environmental readings are checked and normalised, and relevant derived measurements are prepared for analysis.

Layer 3 – Backend Communication

Node.js and Express provide API endpoints. Sensor readings are transmitted over Wi-Fi. Socket.IO supports real-time dashboard updates. MQTT may be introduced for device communication as the system expands.

Layer 4 – Intelligence and Storage

MongoDB stores appropriate sensor readings and user preferences. A rule-based engine analyses workspace conditions. The recommendation service generates explainable suggestions. Future machine-learning modules can support personalisation.

Layer 5 – User Interface and Adaptive Control

The React dashboard displays workspace status. Users configure modes and preferences, accept or reject recommendations, and send approved commands to compatible devices.

Overall data flow: Sensors and webcam → Data processing → Backend API → Recommendation engine → Dashboard and user-approved workspace controls.

13. Methodology

Step 1: Hardware Integration

Connect the ESP8266 to the available sensors and verify each sensor independently.

Step 2: Sensor Data Acquisition

Develop firmware to read temperature, humidity, brightness, proximity and other available measurements.

Step 3: Computer Vision Integration

Use Python and OpenCV to process webcam frames locally and investigate face position and screen-distance estimation.

Step 4: Backend Development

Develop REST APIs using Node.js and Express to receive sensor readings and process user preferences.

Step 5: Database Integration

Use MongoDB to store appropriate measurements, user settings and recommendation history.

Step 6: Recommendation Engine

Develop configurable rules that combine environmental conditions, work-session information and voluntary user feedback.

Step 7: Dashboard Development

Build a React dashboard to display sensor information, recommendations, workspace modes and controls.

Step 8: Adaptive Control

Integrate compatible lighting devices and demonstrate user-approved ambiance changes.

Step 9: Testing and Evaluation

Test individual modules and complete system workflows under different workspace conditions.

14. Privacy, Security and Ethical Considerations

Privacy is a core design requirement of DeskSense AI.

- Opt-in monitoring: Users decide when monitoring is active.
- Local webcam processing: Process webcam frames locally wherever practical.
- No unnecessary video storage: Store derived measurements rather than raw footage.

- No automatic emotion labelling: Do not classify users as sad, angry or stressed based on facial expressions.
- Voluntary feedback: Allow users to report their own comfort or stress levels.
- User-owned information: Provide options to view, export and delete personal records.
- Employer separation: Individual well-being information should not be available to managers for employee performance evaluation.
- Transparent recommendations: Explain why each recommendation was generated.
- Manual override: Allow users to reject, pause or customise any automated action.
- Secure communication: Use authentication, access controls and encrypted communication where supported.

The project must not be used to make employment decisions or to conduct covert psychological surveillance.

15. Implementation Plan

Phase	Main activities	Expected output
Phase 1 – Monitoring	Sensor integration, webcam processing and dashboard	Working workspace monitoring prototype
Phase 2 – Intelligence	Recommendation engine, user feedback and workspace modes	Context-aware recommendation system
Phase 3 – Adaptation	RGB lighting, optional audio and connected-device integration	Adaptive workspace demonstration

This phased approach allows the core prototype to be developed using existing hardware before additional components are purchased.

16. Testing and Evaluation Plan

The following tests are proposed. Their results will be documented after implementation.

Test	Expected behaviour
LDR covered and uncovered	ADC readings should change with light conditions
Temperature changes	Valid temperature readings should be received
Humidity changes	Valid humidity readings should be received
Ultrasonic proximity test	Distance readings should respond to nearby objects
Webcam test	Face position and selected landmarks should be processed
Break reminder test	Reminder should appear after the configured duration
Low-light rule	Appropriate lighting recommendation should appear
User feedback test	Selected user state should influence recommendations
Calm Mode test	User-approved lighting setting should be applied
Privacy test	Raw webcam frames should not be unnecessarily stored
Manual override test	User should be able to reject or stop automation

Evaluation should consider sensor reliability, recommendation correctness, response latency, usability, user acceptance and privacy compliance. No performance or accuracy results are claimed until these tests have been conducted.

17. Limitations

- Facial expressions cannot reliably establish a person's emotional state.
- Webcam-based distance estimation requires calibration and may vary with camera placement.
- The HC-SR04 is not a dedicated face-distance sensor.
- An LDR provides raw light-dependent readings unless calibrated against a suitable reference.
- DHT sensors cannot measure CO_2 or establish air quality.
- Sensor readings alone cannot establish productivity, stress or medical conditions.
- Smart-device automation depends on compatible hardware and communication interfaces.
- Personalised recommendations require user feedback and sufficient evaluation.

These limitations will be addressed progressively through calibration, additional sensing, user feedback and controlled testing.

18. Future Scope

- Integration with dedicated CO_2 and air-quality sensors.
- Advanced computer vision for calibrated ergonomic measurements.
- Personalised recommendation models based on user-approved preferences.
- Integration with compatible smart-home and office devices.
- Optional wearable integration for activity and physiological context.
- Desktop application for focus sessions and notification management.
- Multi-workstation dashboards for facilities management using aggregated, non-identifying environmental information.
- Improved multilingual voice guidance.
- Offline operation and local data processing.
- Expanded accessibility features and configurable user interfaces.

Any future emotion-related analysis must undergo appropriate validation and ethical review before being used in real-world deployments.

19. Expected Outcomes

- A functional IoT-based workspace monitoring prototype.
- A dashboard displaying environmental and workspace information.
- A computer-vision module for selected ergonomic measurements.
- An explainable recommendation engine.
- Personalised Calm, Focus, Refresh and Mindfulness modes.
- A demonstration of user-approved adaptive lighting.
- A privacy-conscious data-management approach.
- A modular architecture suitable for future expansion.

The project aims to demonstrate how affordable IoT hardware and AI-assisted software can work together to create more comfortable and personalised work environments.

20. Conclusion

DeskSense AI proposes a shift from conventional workspace monitoring towards intelligent, adaptive and human-centred workspace assistance.

By combining environmental sensors, computer vision, contextual analysis and voluntary user feedback, the system aims to provide meaningful recommendations that support physical workspace comfort and a more positive working experience.

Its most distinctive aspect is the integration of workspace intelligence with personalised ambiance assistance. Rather than simply reporting measurements, DeskSense attempts to translate them into practical actions and user-controlled environmental adjustments.

With its modular hardware, scalable software architecture and privacy-first approach, DeskSense AI has potential applications in corporate offices, educational institutions, co-working spaces and home-working environments.

The long-term vision is to create a workspace that understands its conditions, respects its user and adapts to individual preferences.

Project Identity

Attribute	Description
Project Name	DeskSense AI
Project Category	Artificial Intelligence, Internet of Things (AIoT), Smart Workspace and Human-Centred Computing
Project Type	Hardware and Software Integrated Prototype
Primary Innovation	Context-aware workspace recommendations combined with personalised, user-approved ambiance adaptation
Tagline	A Workspace That Adapts to You.